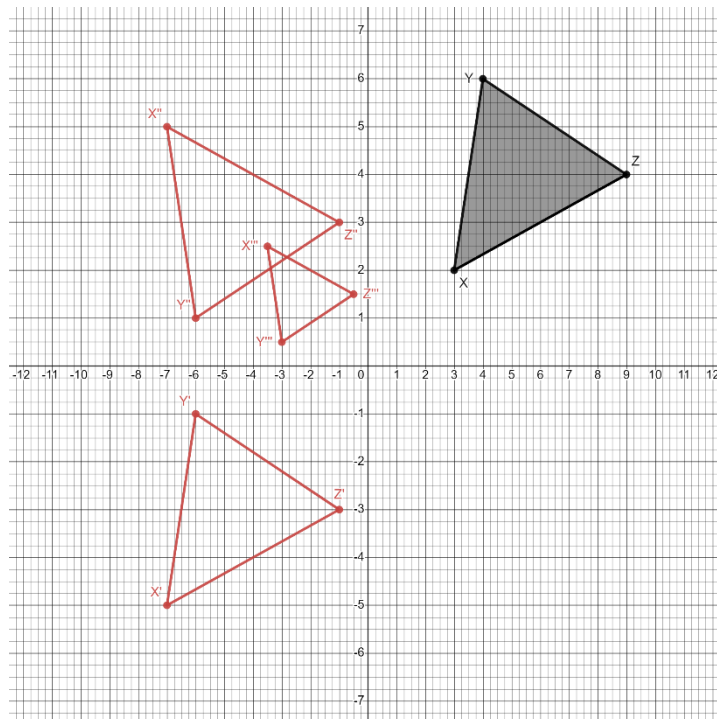


Geometry
Unit 6
Lesson H3 Homework

Name Answer Key
Date _____
Period _____

$\triangle XYZ$ is shown.



Draw the result of each transformation in the following sequence:

1. $\triangle X'Y'Z'$ as a result of $(x, y) \rightarrow (x - 10, y - 7)$
2. $\triangle X''Y''Z''$ as a result of $(x, y) \rightarrow (x, -y)$
3. $\triangle X'''Y'''Z'''$ as a result of $(x, y) \rightarrow \left(\frac{1}{2}x, \frac{1}{2}y\right)$

Indicate the correct relationship for the resulting image:

4. Image $\triangle X'Y'Z'$ is

☒ congruent
☐ similar

 to preimage $\triangle XYZ$.
5. Image $\triangle X''Y''Z''$ is

☒ congruent
☐ similar

 to preimage $\triangle XYZ$.
6. Image $\triangle X'''Y'''Z'''$ is

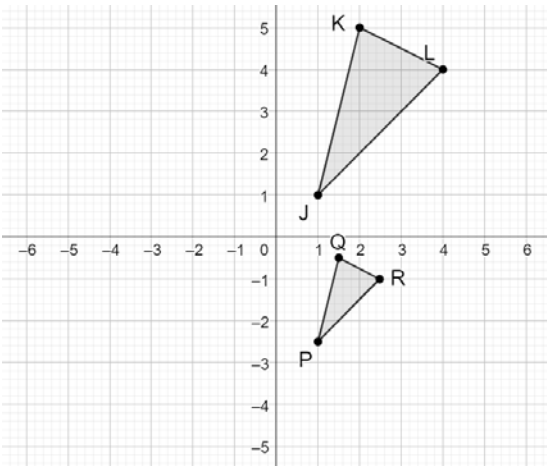
☐ congruent
☒ similar

 to preimage $\triangle XYZ$.

$\triangle JKL$ and $\triangle PQR$ are shown.

Assume: $\frac{JK}{PQ} = \frac{KL}{QR} = \frac{LJ}{RP}$

7. Write a series of transformations that will map $\triangle JKL$ onto $\triangle PQR$ so that $\triangle JKL \sim \triangle PQR$.
 $(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$ then $(x,y) \rightarrow (x + \frac{1}{2}, y - 3)$

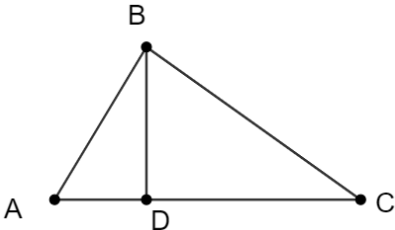


8. If it is given that $\triangle PQR$ is a result of $(x,y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$ then $(x,y) \rightarrow (x + \frac{1}{2}, y - 3)$,
then the triangles are

☐ congruent
☒ similar

.

$\triangle ABC$ is shown with $\overline{BD}$.



Complete the table:

	If Given	True or False? $\triangle ABC \sim \triangle ADB$ by AA Similarity	Justification
9.	$\overline{AB} \perp \overline{BC}$ and $\overline{BD} \perp \overline{AC}$	☒ True ☐ False	$\angle ABC \cong \angle ADB, \angle BAC \cong \angle DAB$
10.	$\frac{AB}{AD} = \frac{BC}{BD}$	☐ True ☒ False	Two sides are proportional, but no angles nor the third side are given in order to confirm similarity.